BANGLADESH: DEBT SUSTAINABILITY ANALYSIS[9]

This attachment summarizes the debt sustainability analysis (DSA) for Bangladesh using the new framework for low-income countries. The macroeconomic scenario underlying this DSA is the program scenario discussed in this staff report. Sensitivity of the profile of the debt-to-GDP ratio to a number of standardized shocks is also discussed.

A. Main Assumptions

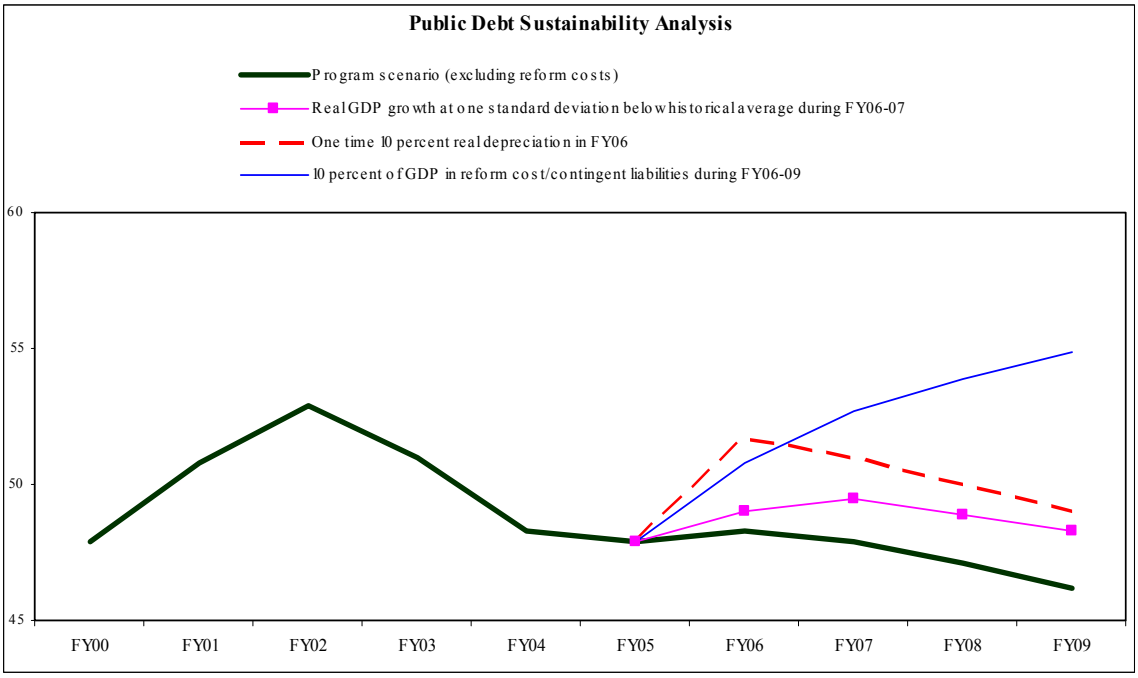
- Economic growth will accelerate to 6½ percent over the medium term, reflecting surging private investment (including FDI) and export diversification.
- Monetary policy is geared toward achieving the government's objective of lowering inflation and maintaining sound external positions. GDP deflator is projected to decline to 4 percent by FY08 and staying at that level thereafter.
- Reflecting efficiency gains from revenue reform, the tax-GDP ratio would rise steadily from 8.2 percent of GDP in FY04 to 9.5 percent of GDP in FY09.
- The increased domestic revenue mobilization, together with sustained external financing, will allow the government to finance pro-poor and pro-growth spending, with a view to achieving the MDGs. The wage bill will be contained through strict limits on wage increases and civil service reform. The budget deficit is projected to stabilize at around 4 percent.
- Structural reforms in NCBs and SOEs will continue to improve governance, efficiency in resource allocation, and investment climate.

B. Total Public Sector Debt Sustainability

The program scenario projects a gradual decline in central government debt, from about 53 percent of GDP in FY02 to about 46 percent of GDP in FY09, due mainly to the reduction in the budget deficit and improved growth prospects. Similarly, the NPV of public debt would also decline from about 34 percent in FY02 to about 30 percent in FY09. To check the robustness of the conclusions, we subject this scenario to a number of stress tests (see Table 1). These include: (i) the key variables kept at their respective historical averages; (ii) a lower long-run GDP growth; (iii) real GDP and primary balance kept below their historical mean; (iv) one-time 10 percent real exchange rate depreciation; (v) 10 percent of GDP in reform costs/contingent liabilities.

[9] As the new procedures on DSAs were only established by the Board after the end of the Article IV consultations, this DSA has not been undertaken jointly with the World Bank.

The sensitivity analyses suggest that the program scenario is most vulnerable to a shock to reform cost/contingent liabilities and a large real depreciation of the exchange rate (Table 2 and Chart below). A one-time real exchange rate depreciation of 10 percent would increase the debt–to-GDP to about 58 percent by FY09. Other macroeconomic shocks analyzed would have an impact of up to 2-percenatge point of GDP on the debt dynamics. The potential restructuring costs of SOE and NCB reforms are tentatively estimated at up to 10 percent of GDP (i.e., below the line in the budget for eventual NCB recapitalization and resolution of SOE arrears), which are assumed to be absorbed during FY06–09. This would have a significant, but transitory impact on the public debt ratio to about 55 percent of GDP by FY09.

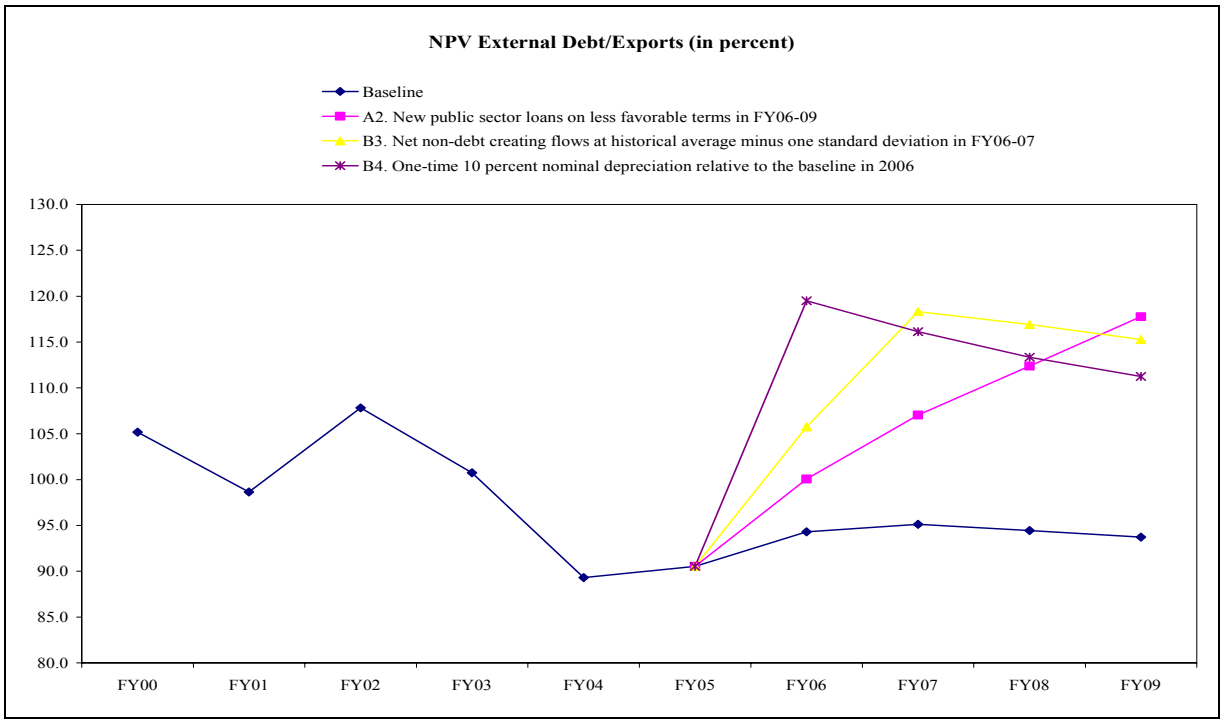


C. External Debt Sustainability

More than half of the total public sector debt analyzed above concerns external debt. Under the program scenario, external debt is roughly stable around 29 percent of GDP, and in NPV terms it is increasing from 90 to about 94 percent of exports between FY05 and FY09, but still remains relatively low, and comfortably below HIPC thresholds. The increase in the NPV-to-exports ratio is effectively caused by the projected post-MFA decline in export growth. Almost all of Bangladesh's external debt is on concessional terms, with an average grant element of 55 percent. At about 7 percent of exports, debt service payments also remain manageable.

Subjecting the baseline scenario to a number of stress tests indicates that the external debt should remain manageable. The ratio of NPV to exports remains below 120 percent in

all cases analyzed. However, a significant increase in NPV ratios does occur in some of the sensitivity analyses (chart below). Contracting of loans on less favorable terms (assuming that the interest rate on new borrowing is 2 percentage points higher than under the baseline scenario) would lead to a steady increase of the NPV ratio to 118 percent of exports in FY09, as opposed to 94 percent under the baseline scenario, confirming the sensitivity of public debt to interest rates. Less nondebt creating flows, such as official and private transfers and FDI, would lead to a similar increase in the NPV ratio, albeit declining at the end of the projection period, showing the important role of continued external inflows. Finally, a one-time 10 percent nominal depreciation would lead to the largest initial shock to the NPV ratio. It would decline thereafter, but remain considerably above the level of the baseline. This supports the argument for keeping exchange-rate depreciations orderly.



Annex V -- Table 1. Bangladesh: External Debt Sustainability Framework 1/

(In percent of GDP, unless otherwise indicated)

	Actual					Historical Average 6/	Standard Deviation 6/	Projections				
	FY00	FY01	FY02	FY03	FY04			FY05	FY06	FY07	FY08	FY09
External debt (nominal) 1/	**32.5**	**33.5**	**34.2**	**31.9**	**29.5**	**...**	**...**	**29.3**	**29.8**	**29.7**	**29.2**	**28.9**
Of which : public and publicly guaranteed (PPG)	32.5	33.5	34.2	31.9	29.5	...	...	29.3	29.8	29.7	29.2	28.9
Change in external debt	-0.1	1.0	0.7	-2.3	-2.4	...	...	-0.2	0.6	-0.1	-0.5	-0.2
Identified net debt-creating flows	-0.9	1.5	-1.6	-3.6	-3.8	...	...	-0.3	-0.2	-0.2	-0.3	-0.1
Noninterest current account deficit	**0.4**	**1.8**	**-0.8**	**-0.5**	**-0.6**	**0.4**	**1.0**	**1.5**	**1.9**	**1.8**	**1.8**	**2.0**
Deficit in balance of goods and services	5.3	6.2	4.8	5.6	5.6	...	...	8.1	9.0	9.1	8.8	8.5
Exports	13.9	15.3	14.3	14.3	14.9	...	...	14.6	14.2	14.0	13.9	13.9
Imports	19.2	21.5	19.0	19.9	20.5	...	...	22.6	23.2	23.1	22.7	22.4
Net current transfers (negative = inflow)	-5.1	-4.6	-5.9	-6.6	-6.6	-5.3	1.0	-6.9	-7.4	-7.5	-7.1	-6.7
Other current account flows (negative = net inflow)	0.1	0.2	0.3	0.6	0.4			0.3	0.3	0.2	0.2	0.2
Net FDI (negative = inflow)	**-0.6**	**-0.7**	**-0.7**	**-0.7**	**-0.7**	**-0.4**	**0.4**	**-0.7**	**-0.7**	**-0.7**	**-0.6**	**-0.6**
Endogenous debt dynamics 2/	**-0.6**	**0.4**	**-0.1**	**-2.4**	**-2.6**	**...**	**...**	**-1.2**	**-1.4**	**-1.4**	**-1.5**	**-1.5**
Contribution from nominal interest rate	0.3	0.3	0.3	0.3	0.3	...	...	0.3	0.3	0.3	0.3	0.3
Contribution from real GDP growth	-1.9	-1.7	-1.5	-1.7	-1.6	...	...	-1.4	-1.7	-1.7	-1.8	-1.8
Contribution from price and exchange rate changes	0.9	1.8	1.1	-1.1	-1.3	...	...	…	…	…	…	…
Residual (3-4) 3/	**0.8**	**-0.6**	**2.4**	**1.3**	**1.4**	**...**	**...**	**0.1**	**0.7**	**0.1**	**-0.2**	**-0.1**
Of which : exceptional financing	0.0	0.0	0.0	0.0	0.0			0.0	-1.2	-1.2	-1.2	-1.3
NPV of external debt 4/	14.6	15.1	15.4	14.4	13.3	...	...	13.2	13.4	13.4	13.1	13.0
In percent of exports	105.2	98.6	107.8	100.7	89.3	...	...	90.5	94.3	95.1	94.4	93.7
NPV of PPG external debt	**14.6**	**15.1**	**15.4**	**14.4**	**13.3**	**...**	**...**	**13.2**	**13.4**	**13.4**	**13.1**	**13.0**
In percent of exports	**105.2**	**98.6**	**107.8**	**100.7**	**89.3**	**...**	**...**	**90.5**	**94.3**	**95.1**	**94.4**	**93.7**
Debt service-to-exports ratio (in percent)	9.2	8.3	8.6	8.1	7.8	...	...	7.0	7.2	7.3	7.3	7.1
PPG debt service-to-exports ratio (in percent)	9.2	8.3	8.6	8.1	7.8	...	...	7.0	7.2	7.3	7.3	7.1
Total gross financing need (billions of U.S. dollars)	0.5	1.1	-0.2	0.0	-0.1	...	...	1.1	1.4	1.5	1.6	1.8
Noninterest current account deficit that stabilizes debt ratio	0.5	0.9	-1.6	1.8	1.8	...	...	1.7	1.3	2.0	2.4	2.2
Key macroeconomic assumptions												
Real GDP growth (in percent)	5.9	5.3	4.4	5.3	5.5	5.1	2.0	5.2	6.0	6.0	6.5	6.5
GDP deflator in US dollar terms (change in percent)	-2.7	-5.2	-3.1	3.3	4.1	-0.8	3.4	1.6	-2.4	0.5	2.0	1.9
Effective interest rate (percent) 5/	1.0	1.0	1.0	1.0	1.0	1.0	0.4	1.1	1.1	1.1	1.1	1.1
Growth of exports of G&S (U.S. dollar terms, in percent)	9.3	9.6	-5.4	8.6	14.4	8.0	38.8	4.8	1.1	5.2	7.5	8.5
Growth of imports of G&S (U.S. dollar terms, in percent)	6.3	11.5	-10.3	13.5	13.2	6.5	38.4	18.3	6.1	6.0	6.5	7.0
Grant element of new public sector borrowing (in percent)	55.0	55.0	55.0	55.0	55.0	55.0	0.0	55.0	55.0	55.0	55.0	55.0
A. Alternative scenarios (NPV external debt/exports, in percent)												
A1. Key variables at their historical averages in FY06-09 7/	...	...	...	...	...	...	...	90.5	89.7	88.8	87.8	86.3
A2. New public sector loans on less favorable terms in FY06-09 8/	...	...	...	...	...	...	...	90.5	100.1	107.0	112.4	117.8
B. Bound tests (NPV external debt/exports, in percent)												
B1. Real GDP growth at historical average minus one standard deviation in FY06-07	...	...	...	...	...	...	...	90.5	96.4	99.3	98.6	97.9
B2. Export value growth at historical average minus one standard deviation in FY06-07 9/	...	...	...	...	...	...	...	90.5	94.9	102.1	101.3	100.5
B3. Net nondebt creating flows at historical average minus one standard deviation in FY06-07 10/	...	...	...	...	...	...	...	90.5	105.7	118.3	116.9	115.3
B4. One-time 10 percent nominal depreciation relative to the baseline in 2006 11/	...	...	...	...	...	...	...	90.5	119.5	116.1	113.3	111.2

Source: Fund staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), noninterest current account in percent of GDP, and nondebt creating flows.
8/ Assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
9/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
10/ Includes official and private transfers and FDI.
11/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

Annex V -- Table 2. Bangladesh: Public Sector Debt Sustainability Framework
(In percent of GDP, unless otherwise indicated)

	Actual					Historical	Standard	Projections				
	FY00	FY01	FY02	FY03	FY04	Average 5/	Deviation 5/	FY05	FY06	FY07	FY08	FY09
Baseline medium-term projections												
Public sector debt 1/	47.9	50.8	52.9	51.0	48.3	...	...	47.9	48.3	47.9	47.1	46.2
Of which : foreign-currency denominated	32.5	33.5	34.2	31.9	29.5	...	...	29.3	29.8	29.7	29.2	28.9
Change in public sector debt	2.1	2.9	2.1	-1.9	-2.7	...	...	-0.4	0.5	-0.4	-0.8	-0.9
Identified debt-creating flows	1.5	5.2	0.0	-2.2	-1.5	...	...	0.2	0.1	-1.0	-1.0	-0.9
Primary deficit	1.8	3.1	1.3	0.4	1.1	...	...	1.8	1.9	1.6	1.4	1.4
Revenue and grants	9.9	10.1	11.1	11.4	10.5	...	...	11.1	11.7	12.0	12.1	12.3
Of which : grants	*1.5*	*1.1*	*1.0*	*1.1*	*0.5*	...	...	0.6	0.7	0.8	0.8	0.7
Primary (noninterest) expenditure	11.6	13.1	12.5	11.8	11.6	...	...	13.0	13.6	13.6	13.6	13.7
Automatic debt dynamics	-0.2	2.1	-1.4	-2.6	-2.6	...	...	-1.6	-1.8	-2.6	-2.4	-2.3
Contribution from interest rate/growth differential	-1.6	-1.4	-1.5	-2.3	-2.3	...	...	-2.1	-2.3	-2.3	-2.4	-2.4
Of which: contribution from average real interest rate	1.0	1.0	0.6	0.4	0.4	...	...	0.3	0.4	0.4	0.5	0.5
Of which: contribution from real growth	-2.6	-2.4	-2.1	-2.7	-2.7	...	...	-2.4	-2.7	-2.7	-2.9	-2.9
Contribution from real exchange rate depreciation	1.4	3.6	0.2	-0.3	-0.3	...	...	0.5	0.5	-0.3	0.0	0.0
Other identified debt-creating flows	0.0	0.0	0.0	0.0	0.0	...	...	0.0	0.0	0.0	0.0	0.0
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	...	...	0.0	0.0	0.0	0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0	0.0	...	...	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	...	...	0.0	0.0	0.0	0.0	0.0
Other (specify, e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	...	...	0.0	0.0	0.0	0.0	0.0
Residual, including asset changes	0.6	-2.3	2.2	0.2	-1.2	...	...	-0.6	0.4	0.6	0.1	0.1
NPV of public sector debt	30.0	32.4	34.1	33.5	32.0	...	...	31.8	31.9	31.6	31.0	30.3
Of which: foreign-currency dominated	14.6	15.1	15.4	14.4	13.3	...	...	13.2	13.4	13.4	13.1	13.0
Of which: external	14.6	15.1	15.4	14.4	13.3	...	...	13.2	13.4	13.4	13.1	13.0
NPV of contingent liabilities (not included in public sector debt)	...	...	...	...	...	...	...	...	...	...	...	...
Gross financing need 2/	4.6	5.0	4.5	3.2	3.4	...	...	4.3	4.2	4.0	3.9	3.8
NPV of public sector debt-to-revenue ratio (in percent) 3/	303.5	321.2	306.4	294.0	303.7	...	...	285.2	273.5	263.8	255.6	246.6
Of which: external	147.8	149.5	138.2	126.2	125.9	...	...	118.4	114.8	111.7	108.2	105.7
Debt service-to-revenue ratio (in percent) 3/ 4/	25.7	25.4	24.2	24.2	21.6	...	...	22.2	20.0	20.1	19.9	19.3
Primary deficit that stabilizes the debt-to-GDP ratio	-0.3	0.2	-0.8	2.4	3.8	...	...	2.2	1.4	2.0	2.3	2.3
Fiscal sustainability	2.1	2.9	2.1	-1.9	-2.7	...	...	-0.4	0.5	-0.4	-0.8	-0.9
Key macroeconomic and fiscal assumptions												
Real GDP growth (in percent)	5.9	5.3	4.4	5.3	5.5	5.1	0.4	5.2	6.0	6.0	6.5	6.5
Average nominal interest rate on forex debt (in percent)	1.1	1.1	1.0	1.0	1.0	1.1	0.1	1.0	1.1	1.1	1.0	1.0
Average real interest rate on domestic currency debt (in percent)	8.2	7.2	5.9	4.6	2.2	2.5	4.7	2.7	2.3	3.3	4.5	4.7
Real exchange rate depreciation (in percent, + indicates depreciation)	4.5	11.6	0.5	-1.0	-1.1	2.0	4.4	1.7	...	...	...	...
GDP deflator in taka terms (change in percent)	1.9	1.6	3.2	4.4	5.5	4.1	1.7	6.0	5.5	4.8	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percent)	20.8	18.8	-0.9	-0.2	4.0	4.5	10.0	17.1	11.1	6.0	6.6	7.5
Grant element of new external borrowing (in percent)	55.0	55.0	55.0	55.0	55.0	55.0	0.0	55.0	55.0	55.0	55.0	55.0
A. Alternative scenarios												
A1.Slow reform Scenario	...	...	...	...	...	...	...	47.9	49.0	48.7	48.2	47.6
A2. Historical average scenario	...	...	...	...	...	...	...	47.9	48.1	47.7	47.3	46.9
A3. Lower long-run GDP growth	...	...	...	...	...	...	...	47.9	48.4	48.1	47.4	46.7
B. Bound tests												
B1. Real GDP growth is at one standard deviation below historical average in each of the first two projection years (FY06-07)	...	...	...	...	...	...	...	47.9	49.0	49.5	48.9	48.3
B2. Primary balance is at one standard deviation below historical average in each of the first two projection years (FY06-07)	...	...	...	...	...	...	...	47.9	48.7	48.8	48.0	47.1
B3. Combination of B1-B3 using one-half a standard deviation below historical average in each of the first two projection yea	...	...	...	...	...	...	...	47.9	48.6	48.8	48.0	47.0
B4. One time 10 percent real depreciation in FY06	...	...	...	...	...	...	...	47.9	51.7	51.0	50.0	49.0
B5. 10 percent of GDP in reform costs/contingent liabilities during FY06-09	...	...	...	...	...	...	...	47.9	50.8	52.7	53.9	54.9

Sources: Country authorities; and Fund staff estimates and projections.

1/ Central government debt on gross basis.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues including grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.